

House Price Prediction — Summary of Findings

A regression analysis of 545 residential properties (12 features: size, rooms, amenities, and furnishing) was used to predict sale price and identify the strongest price drivers.

Model Performance

Model	MAE	RMSE	R ² Score
Linear Regression	970,043	1,324,507	0.653
Random Forest	1,014,947	1,399,769	0.612

Linear Regression slightly outperformed Random Forest, indicating the relationship between features and price is largely linear and additive for this dataset.

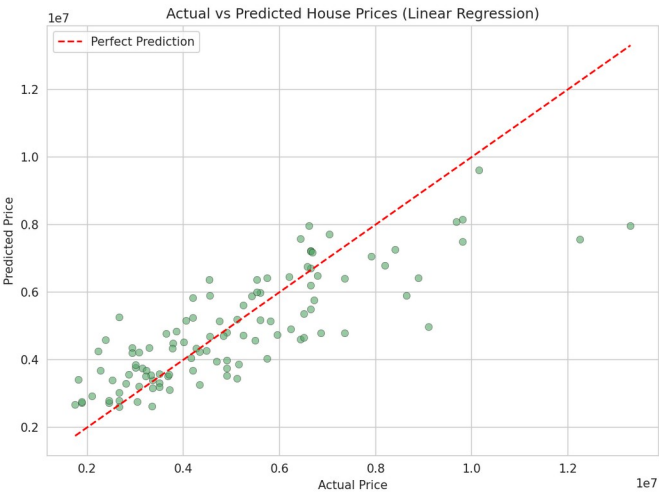
Top Price Drivers

- Area (square footage) — the single strongest predictor of price in both models.
- Number of bathrooms — a much larger effect than bedroom count.
- Air conditioning and hot water heating — amenities with a sizeable positive impact.
- Preferred area location and number of stories also meaningfully raise price.
- Furnishing status — unfurnished homes sell for clearly less than furnished ones.

Insights & Recommendation

The model explains about 65% of price variation ($R^2 \approx 0.65$) with a typical error under 1,000,000 in the dataset's currency — good for ballpark estimates, not exact pricing. A genuine surprise was how little bedroom count mattered compared to bathrooms, area, and amenities.

Recommendation: Real estate agents should anchor pricing guidance and marketing on area, bathroom count, AC/heating, and location desirability — and consider that low-cost upgrades like adding air conditioning or furnishing a unit before sale may yield outsized returns relative to bedroom-focused renovations.



Actual vs. Predicted Price (Linear Regression) — points near the dashed line are accurate predictions.